Auteur: Raj Shah
Studierichting: Informatica
Oefeningenreeks 5G nr. 5.3

$f(x) = \cos x + \sin x$ en $g(x) = \cos x - \sin x$ op interval $\left[0, \frac{\pi}{4}\right]$

$$V = \pi \int_0^{\frac{\pi}{4}} \left((\cos x + \sin x)^2 - (\cos x - \sin x)^2\right) dx$$

$$= \pi \int_0^{\frac{\pi}{4}} \left((\cos^2 x + 2\sin x \cos x + \sin^2 x) - (\cos^2 x - 2\sin x \cos x + \sin^2 x)\right) dx$$

$$= \pi \int_0^{\frac{\pi}{4}} 4 \sin x \cos x dx = 2\pi \int_0^{\frac{\pi}{4}} 2 \sin x \cos x dx$$

$$= 2\pi \int_0^{\frac{\pi}{4}} \sin 2x dx$$

$$= 2\pi \left[\frac{-\cos 2x}{2}\right]_0^{\frac{\pi}{4}} = \frac{2\pi}{2} \cdot \left(-\left(\cos\frac{\pi}{2} - \cos 0\right)\right)$$

$$= \pi \cdot (0 + 1) = \pi$$

References

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